SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1741**

Register Number	**192425396**
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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 10%

QUESTIONS:2 **Total Marks:20**

Descriptive Test

1. **A smart home automation system is being designed to manage lighting, temperature, and security based on user behaviour and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.**

Smart Home Automation Using Intelligent Agents

1. Introduction

A smart home automation system uses intelligent agents to control lighting, temperature, and security automatically. It receives real-time information from sensors such as motion detectors, temperature sensors, and door sensors. Intelligent agents help the system make decisions based on current conditions, user goals, and preferences.

2. Types of Intelligent Agents and Their Applications

1. Simple Reflex Agent

A simple reflex agent makes decisions based only on the current input using predefined condition-action rules. It does not remember past events or learn user preferences.

Application: If a motion sensor detects a person, the agent switches on the lights. If the temperature exceeds a fixed limit, it turns on the air conditioner.

Advantages: It is simple, fast, and suitable for basic automation.

Limitation: It cannot consider past behaviour or make decisions based on changing situations.

2. Model-Based Agent

A model-based agent maintains an internal model of the environment. It uses current sensor information and previous observations to understand the home's state.

Application: If a person leaves a room, the agent checks whether anyone else is present before switching off the lights. It can also consider whether a door is locked and whether the security system is active.

Advantages: It handles changing environments and makes better decisions than a simple reflex agent.

Limitation: Maintaining an accurate model requires additional memory and processing.

3. Goal-Based Agent

A goal-based agent selects actions that help achieve specific goals. It considers possible actions and chooses those that move the system towards its objectives.

Application: If the goal is to save energy, the agent switches off unused lights and adjusts the thermostat when the house is empty. If the goal is security, it locks doors and activates alarms when the residents leave.

Advantages: It can plan actions and handle different situations according to specific goals.

Limitation: Planning may require more time and computation.

4. Utility-Based Agent

A utility-based agent evaluates different possible actions using a utility function and selects the action that provides the greatest overall benefit. It considers factors such as comfort, energy consumption, safety, and user preferences.

Application: On a hot day, the agent selects a comfortable temperature while minimizing electricity usage. At night, it may use dim lighting to save energy while maintaining safety.

Advantages: It balances multiple objectives and can adapt decisions to user preferences.

Limitation: Designing a suitable utility function can be complex, and different objectives may conflict.

3. Comparison of Intelligent Agents

Agent type	Main function	Smart home example
Simple reflex	Responds to current input	Turns on lights when motion is detected
Model-based	Maintains an internal state	Checks room occupancy before switching off lights
Goal-based	Works towards a specific goal	Reduces energy use when the home is empty
Utility-based	Balances multiple benefits	Maintains comfort while reducing electricity costs

4. Most Effective Approach: Hybrid Intelligent Agent

A hybrid approach combining all four types of agents would be most effective for smart home automation.

- Simple reflex agent: Provides immediate responses to motion, temperature, and security sensor inputs.
- Model-based agent: Maintains information about room occupancy, door status, and previous observations.
- Goal-based agent: Plans actions to achieve goals such as energy saving, comfort, and home security.
- Utility-based agent: Balances comfort, electricity consumption, safety, and learned user preferences to choose suitable actions.

For example, when a family leaves home, the system detects that the house is empty, switches off unnecessary lights, adjusts the air conditioner, and activates security features. When the family returns, it restores preferred lighting and temperature settings.

Security-related actions should have high priority, so energy-saving decisions never override essential safety requirements.

5. Conclusion

Intelligent agents improve smart home automation by making decisions automatically and efficiently. Simple reflex agents provide quick responses, model-based agents understand the environment, goal-based agents plan actions, and utility-based agents balance competing objectives. Therefore, a hybrid intelligent agent system is the most suitable approach for achieving comfort, energy efficiency, security, and personalized user experiences.

2. **A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.**

1. Introduction

A robot placed in an unknown maze must find a path from its starting position to the exit without knowing the maze layout in advance. Uninformed search algorithms explore possible paths without using additional knowledge about which direction leads to the goal. Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help the robot find a suitable path.

2. Uninformed Search Strategies

A. Uniform Cost Search (UCS)

Uniform Cost Search expands the path with the lowest total cost first. It uses a priority queue to select the next position based on the cost of reaching it.

Working:

1. Start from the robot's initial position.
2. Explore available neighbouring cells.
3. Calculate the total cost of reaching each cell.
4. Expand the cell with the lowest path cost.
5. Continue until the exit is reached.

Example: If one path contains fewer costly movements than another, UCS selects the path with the lower total cost, even if it contains more steps.

Advantages:

- Finds the least-cost path when step costs are non-negative.
- Complete when each step has a positive minimum cost.

Disadvantages:

- Can consume large amounts of memory.
- May explore many unnecessary paths before reaching the exit.

B. Iterative Deepening Search (IDS)

Iterative Deepening Search repeatedly performs depth-limited search, increasing the depth limit each time until the exit is found.

Working:

1. Start searching with a depth limit of 0.
2. Increase the limit to 1, 2, 3, and so on.
3. Explore paths up to the current limit.
4. Repeat the search with a larger limit until the exit is found.

Example: If the exit is five steps away, IDS gradually increases its depth limit until it reaches the exit.

Advantages:

- Requires less memory than UCS.
- Finds the shallowest solution when all actions have equal cost.
- Complete when the branching factor is finite.

Disadvantages:

- Repeats searches from the beginning at each depth.
- Does not guarantee the least-cost path when movement costs differ.

3. Time and Space Complexity

Let \(b\) be the branching factor, \(d\) the depth of the shallowest solution, and \(C^*\) the cost of the optimal solution. For UCS, let \(\epsilon\) be the minimum positive step cost.

Factor	**UCS**	**IDS**
Time complexity	Exponential in the worst case; commonly expressed as \(O(b^{1+\lfloor C^*/\epsilon\rfloor})\)	\(O(b^d)\)
Space complexity	Exponential; stores many frontier and explored nodes	\(O(bd)\)
Completeness	Yes, with a positive minimum step cost	Yes, with finite branching
Optimality	Yes, for non-negative step costs under standard assumptions	Yes, when all step costs are equal
Main benefit	Finds the least-cost path	Uses less memory

Analysis: UCS is suitable when different paths have different movement costs, while IDS is preferable when memory is limited and every movement has the same cost.

4. Relation to Types of Intelligent Agents

1. Simple Reflex Agent

It chooses actions using the current sensor input and predefined rules.

Maze application: If a wall is detected ahead, the robot turns left or right.

Limitation: It cannot remember explored paths and may repeatedly visit the same locations.

2. Model-Based Agent

It maintains an internal representation of the maze using previous observations.

Maze application: It records visited cells, detected walls, and available routes to avoid repeated exploration.

Limitation: Its model may initially be incomplete because the maze is unknown.

3. Goal-Based Agent

It chooses actions to achieve a specific goal, such as reaching the exit.

Maze application: It explores possible routes and selects actions that help it reach the exit.

Limitation: It may require considerable search time in a large maze.

4. Utility-Based Agent

It evaluates possible actions according to their overall benefit, such as path length, energy consumption, time, and safety.

Maze application: It chooses a route that balances short travel distance with lower energy usage or reduced risk.

Limitation: Defining suitable utility values can be difficult.

5. Most Effective Combination

The most effective solution is a model-based, goal-based robot using Uniform Cost Search, especially when the maze has different movement costs.

- The model-based agent stores information about visited cells and discovered walls.
- The goal-based agent directs the search towards the exit.
- UCS selects the lowest-cost route among the known alternatives.
- The robot explores the maze using sensors and updates its internal model whenever new information is discovered.

If all movements have equal cost and memory is limited, IDS is a reasonable alternative because it finds the shallowest solution while using less memory.

Since the maze is initially unknown, the robot cannot plan using a complete map from the beginning. It must discover the environment incrementally and replan when necessary. If a discovered path is blocked, it updates its model and searches again.

6. Conclusion

UCS and IDS are useful uninformed search strategies for solving maze problems. UCS is appropriate for finding the least-cost route, whereas IDS is useful when memory efficiency is important and movement costs are equal. Combining a model-based agent with goal-based decision-making and UCS provides an effective solution for an unknown maze with varying movement costs. This approach supports systematic exploration, informed decisions based on discovered information, and reliable navigation towards the exit.

Code of Conduct:

I D.N Shirshika (192425396 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *D.N Shirshika*

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	5
Explanation	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	5
Application	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	4
Organization	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	3
Accuracy	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps	2
Clarity	Clear, neat, professional	Understandable with minor	Limited clarity, readability	Poorly written, difficult to	1

	presentation	issues	issues	understand	